

MID

1

Mean: Mean normally refers to the average value of a set of numbers. There are three types of Mean.

(i) Arithmetic Mean: The arithmetic mean is the most common type of average. It is calculated by adding up all the numbers in a set and then dividing by the number of values in that set.

Arithmetic Mean for Grouped Data

Formula: $\bar{x} = \frac{\sum f \cdot x}{\sum f}$

Here,

① $\bar{x}$ = arithmetic mean

f = frequency of each class

x = class midpoint

$\sum f$ = total frequency

$\sum f \cdot x$ = sum of all $f \cdot x$.

Example 1:

Class Interval	Frequency (f)	Midpoint (x)	f · x
0-10	5	5	25
10-20	8	15	120
20-30	12	25	300
30-40	5	35	175
$\sum f = 30$			$\sum f \cdot x = 620$

Mean, $\bar{x} = \frac{620}{30} = 20.67$

(ii) Harmonic Mean:

⊗ Assume Mean: Sometimes large values of the variable are involved in calculation of mean. In order to make our computation easier, we can may assume one of the values as the mean.

$$\bar{x} = A + \frac{\sum fd}{n}$$

Where A = Assumed mean $\left[\frac{n}{2} \text{ from the frequency column} \right]$

d = deviation of x

$$d = x - A$$

Example 1:

Waiting time	1.5-2.0	2.0-2.5	2.5-3.0	3.0-3.5	3.5-4.0	4.0-4.5
No. of customers	3	10	18	10	7	2

Solution:

Class Interval	No. of customers, f	Mid value, x	$d = x - A$	fd
1.5-2.0	3	1.75	-1	-3
2.0-2.5	10	2.25	-0.5	-5
2.5-3.0	18	2.75 (A)	0	0
3.0-3.5	10	3.25	0.5	5
3.5-4.0	7	3.75	1	7
4.0-4.5	2	4.25	1.5	3
	$n = 50$			$\sum fd = 7$

$$\text{Assum mean } \bar{x} = A + \frac{\sum fd}{n}$$

$$= 2.75 + \frac{7}{50} = 2.89$$

Geometric Mean: The Geometric Mean is an analytical method of finding the average rate of growth or decline in the values of an item over a particular period of time.

For Grouped data: If f_i is the frequency of x_i then

$$GM = \sqrt[n]{f_1 x_1 * f_2 x_2 * \dots * f_i x_i}$$

Example: The inflation rate in a country in successive year was 1.05, 1.08, 1.12, 1.25 and 1.34. What is the average rate of inflation per year?

Solⁿ: $n = 5$

$$G.M = \sqrt[5]{1.05 \times 1.08 \times 1.12 \times 1.25 \times 1.34}$$

$$= \sqrt[5]{2.127384} = 1.161$$

Harmonic Mean: The Harmonic Mean of a set of numbers is the reciprocal of the arithmetic mean of the reciprocals of the numbers.

For grouped data: If x_i has the frequency f_i then

$$H.M = \frac{n}{\sum \left(\frac{f_i}{x_i} \right)}$$

$$F = 675$$

$$\frac{675}{n} + A = \bar{x}$$

$$68.5 = \frac{7}{0.2} + 28.5 =$$

Example:

Class Interval	Frequency (f_i)	Midpoint (x_i)	$\frac{f_i}{x_i}$
10-20	4	15	0.267
20-30	6	25	0.240
30-40	10	35	0.286
40-50	5	45	0.111
	$\sum f_i = 25$		$\sum (\frac{f_i}{x_i}) = 0.904$

$$H.M = \frac{\sum f_i x_i}{\sum (\frac{f_i}{x_i})} = \frac{25}{0.904} = 27.65$$

Median : Median is the value of the variable that divides the distribution into two equal parts when the values are arranged in order of magnitude.

For grouped data: $Median = L + \frac{\frac{n}{2} - cfb}{f} \times h$

Soln:

Class Interval	Frequency (f)	Cumulative Frequency (cfb)
118-126	8	8
127-135	10	18
136-144	14	32 (cfb)
145-153	9	50
154-162	7	59
163-171	7	66
172-180	4	70
	$n = 76$	

Here, $\frac{n}{2} = \frac{70}{2} = 35$

Class Interval	Frequency	Cumulative Frequency
0-10	21	21
10-20	22	43
20-30	28	71
30-40	2	73
40-50	2	75

$L = 144.5$
 $n = 70$
 $f = 18$
 $h = 9$

Median $= 144.5 + \frac{35 - 32}{18} \times 9$
 $= 144.5 + 1.5$
 $= 146$

Mode: The mode is the value of the data which occurs most frequently. A set of data may have no, one, two or more modes.

For grouped data: $\text{Mode} = L + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \times h$

where, L = Lower class boundary of the modal class

Δ_1 = Difference between the frequency of the modal class and preceding class

Δ_2 = Difference between the frequency of the modal class and subsequent class

h = class width or size.

Ex: For the table below find the mode

Class Interval	11-20	21-30	31-40	41-50	51-60	61-70
Frequency	6	20	12	10	9	9

Class Interval	Frequency
11-20	6
21-30	20
31-40	12
41-50	10
51-60	9
61-70	9

$$n = 66$$

Here, the 2nd class has the highest frequency, hence the modal class is the 2nd class

$$\text{Here, } L = 20.5$$

$$\Delta_1 = 20 - 6 = 14$$

$$\Delta_2 = 20 - 12 = 8$$

$$h = 10$$

$$\text{Mode} = L + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \times h$$

$$= 20.5 + \left(\frac{14}{14 + 8} \right) \times 10$$

$$= 26.9$$

Standard Deviation:

Formula :

$$\sigma = \sqrt{\frac{\sum f(x - \bar{x})^2}{n}}$$

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Example : Compute standard deviation

Ages in years	50-54	55-59	60-64	65-69	70-74	75-79	80-84
No. of people	1	2	10	12	18	25	9

Soln :

Class interval	Frequency f	Mid-value x	fx	$x - \bar{x}$	$f(x - \bar{x})^2$
50-54	1	52	52	-20.06	402.40
55-59	2	57	114	-15.06	453.61
60-64	10	62	620	-10.06	1012.04
65-69	12	67	804	-5.06	307.24
70-74	18	72	1296	-0.06	0.07
75-79	25	77	1925	4.94	610.09
80-84	9	82	738	9.94	889.23
	$n = 77$		$\sum fx = 5549$		$\sum f(x - \bar{x})^2 = 3674.68$

Here,
Mean,

$$\bar{x} = \frac{\sum fx}{n}$$

$$= \frac{5549}{77}$$

$$= 72.06$$

Standard deviation,

$$\sigma = \sqrt{\frac{\sum f(x - \bar{x})^2}{n}}$$

$$= \sqrt{\frac{3674.68}{77}} = \sqrt{47.7231}$$

$$= 6.9082 \text{ (Ans.)}$$

☐ Quartile Deviation (QD) Definition:

Quartile Deviation measures the 'spread' of the middle 50% of data. It's half the difference between the third quartile (Q_3) and the first quartile (Q_1):

$$QD = \frac{Q_3 - Q_1}{2}$$



For,

Ungrouped data, $QD = \frac{Q_3 - Q_1}{2}$: $Q = \text{Position}$

Here; For odd numbers

$$Q_1 = \frac{n+1}{4}$$

$$Q_3 = \frac{n+1}{4} \times 3$$

For Even data numbers

$$Q_1 = \frac{n}{4}$$

$$Q_3 = \frac{n}{4} + 1$$

Grouped data:

$$\text{For grouped data, } Q_3 \approx \frac{3N}{4}$$

$$Q_1 = L_1 + \left(\frac{\frac{3N}{4} - F}{f} \right) \times h$$

f = highest frequency

h = class difference

L_1 = Highest frequency or lower limit

N = total frequency

F = Cumulative frequency of lower limit or

L_3 = highest frequency or lower limit

Definition all (4, 8) (6, 9) &

Pic of Phn (4, 6) Determine correlation

Discuss (var1 / var2)

4(a) → all definition

4(b)

Determine correlation

Correlation: The statistical tool with the help of which this relationships between two or more than two variables is called correlation. The measure of correlation called the efficient of co-relation (denoted by the symbol r) summarize on are figure the direction and degree of correlation. Thus correlation analysis refers to the techniques used in measuring the closeness of the relationship between variables.

AM to define as, "An Analysis of the covariation of two or more variables is usually called correlation."

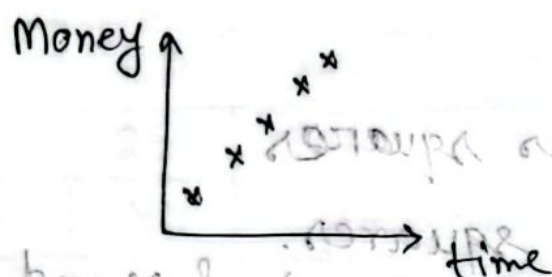
The problem analysis the relation between different series should be broken down into these steps

1. Determining whether a relation exists and if it does measuring it.
2. Rating whether it is significant and
3. Examining the cause and effect if any.

(i) Negative Correlation :

As one variable increases, the other decreases.

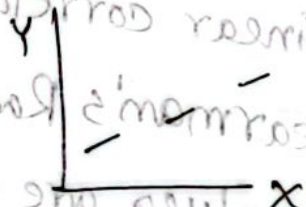
Example : As time increases, money decreases.



(ii) Positive / Exponential Correlation

As one variable increases, the other also increases,

Sometimes at an increasing rate.



(a) Discuss (methods)

Karl Pearson's Coefficient of Correlation :

Definition : Karl Pearson's method measures the linear relationship between two variables. It is suitable for quantitative data and gives a value (r) between -1 and $+1$.

$$\text{Formula : } r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \cdot \sum (y - \bar{y})^2}}$$

$$\text{Shortcut Formula : } r = \frac{N \sum XY - (\sum X)(\sum Y)}{\sqrt{[N \sum X^2 - (\sum X)^2][N \sum Y^2 - (\sum Y)^2]}}$$

r = Pearson's correlation coefficient (result value: -1 to +1)

N = Number of data

$\sum X$ = Sum of all X value

$\sum Y$ = Sum of all Y value

$\sum X^2$ = Sum of all X values squares

$\sum Y^2$ = Sum of all Y values squares.

$\sum XY$ = Sum of the product of each pair of X and Y

if $r = +1 \rightarrow$ Perfect positive correlation

$r = -1 \rightarrow$ Perfect negative correlation

$r = 0 \rightarrow$ no linear correlation.

Rank Correlation (Spearman's Rank Method)

Definition: Used when actual values are not available, only ranks. It measures the degree of agreement between two sets of rankings.

Formula: $R = 1 - \frac{\sum d^2}{n(n^2 - 1)}$

d = difference between ranks

n = number of observations

$$\frac{(\sum Y - Y)(\sum X - X)}{\sqrt{(\sum Y - Y)(\sum X - X)}} = r$$

$$\frac{(\sum Y^2)(\sum X^2) - YX \sum N}{[\sum(Y^2) - Y \sum N][\sum(X^2) - X \sum N]} = r$$

Math of two methods

Pearson's

X	Y	X ²	Y ²	XY
10	20	100	400	200
20	25	400	625	500
30	40	900	1600	1200
40	150	1600	22500	6000
50	155	2500	3025	7750
$\Sigma X = 150$	$\Sigma Y = 190$	$\Sigma X^2 = 6500$	$\Sigma Y^2 = 8150$	$\Sigma XY = 6650$

Now plug values into the shortcut formula:

$$\begin{aligned}
 r &= \frac{N \Sigma XY - (\Sigma X)(\Sigma Y)}{\sqrt{[N \Sigma X^2 - (\Sigma X)^2][N \Sigma Y^2 - (\Sigma Y)^2]}} \\
 &= \frac{5(6650) - (150)(190)}{\sqrt{[5(6500) - 150^2][5(8150) - 190^2]}} \\
 &= \frac{33250 - 28500}{\sqrt{(32500 - 22500)(40750 - 36100)}} \\
 &= \frac{4750}{\sqrt{10000 \times 4650}} = \frac{4750}{\sqrt{46500000}} \approx 0.6969
 \end{aligned}$$

$r \approx 0.70$, there is a moderate to strong positive correlation between math and science marks.

Spearman Rank Correlation (Math)

Example :

Student	Rank in English	In history	$d = R_1 - R_2$	d^2
A	1	2	-1	1
B	2	1	1	1
C	3	3	0	0
D	4	5	-1	1
E	5	4	1	1
				4

Now, $R = 1 - \frac{6 \sum d^2}{N^3 - N}$

$$= 1 - \frac{6 \times 4}{5^3 - 5}$$

$$= 0.8$$

$R = 0.8$ means there is strong positive correlation between the ranks.

$$\frac{[1001 - (0218)2][1001 - (0218)2]}{[1001 - (0218)2][1001 - (0218)2]}$$

$$00288 - 02555$$

$$(1001 - 0218)(1001 - 0218)$$

$$0.8 = \frac{0255}{1001}$$

criticism marks points of storehouse a 21 result, 0.50 = r
A more serious point about correlation is that it is not a measure of causation.

Group C

(b) Problem solve in Regression analysis

Regression Equation of y on x : $y = a + bx$

$$\begin{aligned} \square \sum y &= Na + b \sum x \\ \square \sum xy &= a \sum x + b \sum x^2 \end{aligned}$$

Regression Equation of x on y : $x = a + by$

$$\begin{aligned} \square \sum x &= Na + b \sum y \\ \square \sum xy &= a \sum y + b \sum y^2 \end{aligned}$$

Example: Calculate the regression equations of x on y and y on x from the follow data.

$x = 1 \ 2 \ 3 \ 4 \ 5$

$y = 2 \ 5 \ 3 \ 8 \ 7$

Soln:

x	y	x^2	y^2	xy
1	2	1	4	2
2	5	4	25	10
3	3	9	9	9
4	8	16	64	32
5	7	25	49	35
$\sum x = 15$	$\sum y = 25$	$\sum x^2 = 55$	$\sum y^2 = 151$	$\sum xy = 88$

Here, $n = 5$

$$\sum x = na + b \sum y$$

$$\sum xy = a \sum y + b \sum y^2$$

$$\therefore \sum x = na + b \sum y$$

$$\Rightarrow 15 = 5a + 25b \quad \text{--- (i)}$$

Then, $\sum xy = a \sum y + b \sum y^2$

$$\Rightarrow 88 = 25a + 151b \quad \text{--- (ii)}$$

Now, $\sum y = na + b \sum x$

$$\Rightarrow 25 = 5a + 15b \quad \text{--- (iii)}$$

Then, $\sum xy = a \sum x + b \sum x^2$

$$\Rightarrow 88 = 15a + 55b \quad \text{--- (iv)}$$

equation (i) $\times 5$ and minus with equation (ii) then we get,

$$75 = 25a + 125b$$

$$- 88 = 25a + 151b$$

$$- 13 = 0 - 26b$$

$$\Rightarrow b = 0.5$$

b এর মান (i) এ বসাই,

$$15 = 5a + 25b$$

$$\Rightarrow 15 = 5a + 25 \times 0.5$$

$$\Rightarrow 5a = 2.5$$

$$\therefore a = 0.5$$

regression equation of Y on X :
 $\hat{y} = a + b \hat{x}$

regression equation of X on Y :
 $\hat{x} = a + b \hat{y}$

Example: Calculate the regression equations of X on Y and Y on X from the following data.

X	Y	X ²	Y ²	XY
1	2	1	4	2
2	3	4	9	6
3	4	9	16	12
4	5	16	25	20
5	6	25	36	30
6	7	36	49	42
7	8	49	64	56
8	9	64	81	72
9	10	81	100	90
10	11	100	121	110
11	12	121	144	132
12	13	144	169	156
13	14	169	196	182
14	15	196	225	210
15	16	225	256	240
16	17	256	289	272
17	18	289	324	306
18	19	324	361	342
19	20	361	400	380
20	21	400	441	420
21	22	441	484	462
22	23	484	529	506
23	24	529	576	552
24	25	576	625	600
25	26	625	676	650
26	27	676	729	702
27	28	729	784	756
28	29	784	841	812
29	30	841	900	870
30	31	900	961	930
31	32	961	1024	992
32	33	1024	1089	1056
33	34	1089	1156	1122
34	35	1156	1225	1190
35	36	1225	1296	1260
36	37	1296	1369	1332
37	38	1369	1444	1406
38	39	1444	1521	1482
39	40	1521	1600	1560
40	41	1600	1681	1640
41	42	1681	1764	1722
42	43	1764	1849	1806
43	44	1849	1936	1892
44	45	1936	2025	1980
45	46	2025	2116	2070
46	47	2116	2209	2162
47	48	2209	2304	2256
48	49	2304	2401	2352
49	50	2401	2500	2450
50	51	2500	2601	2550
51	52	2601	2704	2652
52	53	2704	2809	2756
53	54	2809	2916	2862
54	55	2916	3025	2970
55	56	3025	3136	3080
56	57	3136	3249	3192
57	58	3249	3364	3306
58	59	3364	3481	3422
59	60	3481	3600	3540
60	61	3600	3721	3660
61	62	3721	3844	3782
62	63	3844	3969	3906
63	64	3969	4096	4032
64	65	4096	4225	4160
65	66	4225	4356	4290
66	67	4356	4489	4422
67	68	4489	4624	4556
68	69	4624	4761	4692
69	70	4761	4900	4830
70	71	4900	5041	4970
71	72	5041	5184	5112
72	73	5184	5329	5256
73	74	5329	5476	5402
74	75	5476	5625	5550
75	76	5625	5776	5700
76	77	5776	5929	5852
77	78	5929	6084	6006
78	79	6084	6241	6162
79	80	6241	6400	6320
80	81	6400	6561	6480
81	82	6561	6724	6642
82	83	6724	6889	6806
83	84	6889	7056	6972
84	85	7056	7225	7140
85	86	7225	7396	7310
86	87	7396	7569	7482
87	88	7569	7744	7656
88	89	7744	7921	7832
89	90	7921	8100	8010
90	91	8100	8281	8190
91	92	8281	8464	8372
92	93	8464	8649	8556
93	94	8649	8836	8742
94	95	8836	9025	8930
95	96	9025	9216	9120
96	97	9216	9409	9312
97	98	9409	9604	9506
98	99	9604	9801	9702
99	100	9801	10000	9900

Here, n = 100

Equation (iii) $\times 5$ and minus with equation (iv) we get,

$$25 = 15a + 45b$$

$$\begin{array}{r} (-) 25 = 15a + 55b \\ \hline \end{array}$$

$$-13 = -10b$$

$$\Rightarrow b = 1.3$$

(iii) a এর মান বের করি,

$$25 = 5a + 15b$$

$$\Rightarrow 25 = 5a + 15 \times 1.3$$

$$\Rightarrow a = \frac{5.5}{5}$$

$$\therefore a = 1.1$$

Now, Y on X given by,

$$Y = a + bX$$

$$\Rightarrow Y = 1.1 + 1.3X \quad \text{①}$$

The eqⁿ are, $X = 0.5 + 0.5Y$.

$$Y = 1.1 + 1.3X.$$